MODULE 36 · SECTION 5 - MAINTAINING AND AUTOMATING DATA WORKLOADS

Monitoring and Troubleshooting

Cloud Monitoring and alerting, Cloud Logging and sinks, audit logs, BigQuery's monitoring charts and jobs views, and reading quota, resource and access errors.

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Metrics tell you something is wrong. Logs tell you what. Audit logs tell you who.

Where to look for what

QUESTION	LOOK IN
Is something wrong right now?	**Cloud Monitoring** metrics and alerts
What exactly failed, and why?	**Cloud Logging**, and job error messages
Who changed or read this?	**Cloud Audit Logs**
Which BigQuery jobs cost or fail most?	**INFORMATION_SCHEMA.JOBS** and admin charts
What does it cost?	**Billing export**, from module 34

THE PATTERN

Each question has a home. Questions that look in the wrong place are designed as traps.

Alerting policies

- **Metric threshold**: fire when a metric crosses a line for a set window.
- **Metric absence**: fire when a metric stops arriving, such as a silent pipeline.
- **Log-based** policies fire when a particular message appears in the logs.
- Notify through **channels**: email, chat, paging tools, Pub/Sub, webhooks.

> REMEMBER
>
> Put the runbook link in the policy's documentation field, so the alert says what to do.

Metrics worth an alert at Larkspur

SERVICE	METRIC	WHY
Pub/Sub	Oldest unacked message age	Subscribers falling behind
Dataflow	Data freshness, system lag	Streaming job stuck or slow
BigQuery	Slot usage near the reservation maximum	Queries queuing
Bigtable	CPU utilisation of the hottest node	Hotspot or undersized cluster
Managed Airflow	Failed task count, DAG run duration	Nightly run at risk

THE TIEBREAKER

Alert on symptoms users feel, such as freshness, before causes such as CPU.

Routing logs with sinks

- The **Log Router** sends each entry through sinks, each with an inclusion filter.
- Destinations: log buckets, **BigQuery**, Cloud Storage, **Pub/Sub**, another project.
- **Aggregated sinks** collect a whole organisation or folder in one place.
- **Exclusion filters** drop noisy logs. Sinks never route logs from before they existed.

> THE TIEBREAKER
>
> SQL over logs: route to BigQuery, or use Log Analytics with a linked dataset. Third-party tools: Pub/Sub.

Every BigQuery audit log, into one dataset

```
gcloud logging sinks create lk-bq-audit \
  bigquery.googleapis.com/projects/lk-sec-prod/datasets/audit_logs \
  --organization=123456789012 --include-children \
  --log-filter='protoPayload.serviceName="bigquery.googleapis.com"' \
  --use-partitioned-tables
```

3 An aggregated sink: every project in the organisation.

4-5 BigQuery audit entries only, into partitioned tables.

THE TRAP

The sink's service account needs write access on the dataset, or nothing arrives.

Four kinds of audit log

TYPE	RECORDS	ON BY DEFAULT?
Admin Activity	Changes to resources and settings	Always, cannot be disabled
Data Access	Reads of data and metadata	Off, except for BigQuery
System Event	Changes made by Google systems	Always, cannot be disabled
Policy Denied	Access blocked by a security policy	Yes, can be excluded

THE TRAP

Reading Data Access logs needs the Private Logs Viewer role. Logs Viewer alone sees nothing there.

Who deleted the bills table?

```
1 SELECT timestamp,
2        protopayload_auditlog.authenticationInfo.principalEmail AS who,
3        protopayload_auditlog.methodName AS action
4 FROM `lk-sec-prod.audit_logs.cloudaudit_googleapis_com_activity`
5 WHERE protopayload_auditlog.resourceName LIKE '%tables/bills'
6   AND protopayload_auditlog.methodName LIKE '%Delete%'
7   AND DATE(timestamp) >= '2026-10-01'
8 ORDER BY timestamp DESC;
```

2 The principal who made the call.

4 Admin Activity entries land in the activity table.

7 A date filter prunes the partitioned log table.

REMEMBER

Recovery is time travel, from module 23. The audit log only tells you who, and when.

BigQuery's own monitoring

- **Resource utilisation** charts: slots, concurrency, failed jobs, bytes, shuffle, 30 days back.
- The **operational health** dashboard: top active queries and insights, for reservations.
- Both are built on **INFORMATION_SCHEMA**: JOBS, JOBS_TIMELINE and others.
- Failed jobs group by **error type**, so the commonest error stands out.

> HONEST LIMIT
>
> The operational health dashboard does not cover on-demand workloads. Use the utilisation charts there.

BigQuery errors, decoded

ERROR	MEANS	DO
accessDenied 403	No permission on the resource	Grant access. Retrying will not help
rateLimitExceeded 403	Too many requests too quickly	Back off, then retry in seconds
quotaExceeded 403	A longer quota is used up	Read which one. Wait, or redesign
resourcesExceeded 400	One query needs too much	Reshape it: ORDER BY, skew, size
notFound 404	Wrong name, or wrong location	Check names and the job location
backendError 5xx	Temporary service problem	Retry with exponential backoff

THE TRAP

Three different 403s, three different fixes. Read the reason, not just the code.

Troubleshooting, in order

01
Alert
A symptom metric fires, with a runbook link.

02
Scope
Dashboards: what, where, since when?

03
Logs
The error reason and the failing step.

04
Changes
Audit logs: did anyone change something?

05
Fix and learn
Fix, then add the missing alert or test.

THE PATTERN

Most incidents follow a change. Check the audit log before guessing.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
Know when a pipeline goes **silent**	Metric-absence alerting policy
Who deleted or changed it?	Admin Activity audit logs
Who **read** the customer table?	Data Access logs, Private Logs Viewer
Logs from **every project**, queried in SQL	Aggregated sink to BigQuery
Send logs to a **third-party** tool	Sink to Pub/Sub
rateLimitExceeded on bursts	Exponential backoff and retry

THE PATTERN

Something wrong, what, or who: metrics, logs, or audit logs.

> Last Tuesday, Larkspur's head-end stopped publishing meter readings for four hours, and nobody noticed until morning.
>
> There were no errors, because nothing was running at all.

What alert would have caught it?

A. A log-based alert on error entries from the publisher

B. A threshold alert on the subscription's backlog growing

C. A metric-absence alert on the topic's published messages

D. A weekly report of message counts, emailed to the whole team

◆ PAUSE AND DECIDE

What alert would have caught it?

A Log-based errors

Nothing was running, so no errors were logged.

B Backlog growing

No messages arrive, so the backlog never grows.

C Metric absence

Fires when published messages stop arriving.

D Weekly report

Finds out days later.

THE SIGNAL

A silent failure, with no errors: a metric-absence condition.

> Someone deleted a curated BigQuery table overnight. Data Access logging was never changed from its defaults.
>
> The security team must find out who did it.

Where should they look?

A The BigQuery job history of every analyst in the project

B Cloud Monitoring's chart of the dataset's table count

C The project's billing export, filtered to BigQuery

D Admin Activity audit logs for the table's delete call

◆ PAUSE AND DECIDE

Where should they look?

A Job history

Deletes may not run as query jobs, and it is slow to search.

B Table count chart

Shows that it happened, not who did it.

C Billing export

Costs, not actions or identities.

D Admin Activity logs

Always written, and record the principal.

THE SIGNAL

Who changed or deleted a resource: Admin Activity audit logs.

> Larkspur's security team wants every BigQuery audit log entry from all 60 projects in one place, queryable with SQL,
>
> including projects created in future.

What should the team set up?

A An aggregated sink at the organisation, to a BigQuery dataset

B A sink in each of the 60 projects, each to a BigQuery dataset

C A sink at the organisation, routing all the logs to Pub/Sub

D Export each project's logs to CSV in Cloud Storage every night

◆ PAUSE AND DECIDE

What should the team set up?

A Aggregated sink to BigQuery
One sink, every project, new ones included, in SQL.

B Sixty sinks
Works today, misses new projects tomorrow.

C Pub/Sub
For streaming to tools, not SQL by itself.

D Nightly CSV
Manual, late, and not queryable directly.

THE SIGNAL

Every project, including new ones, in SQL: an aggregated sink to BigQuery.

> A loader script streams readings into BigQuery in short, heavy bursts.
>
> Some calls fail with rateLimitExceeded, and others succeed a second later. The script gives up on the first error.

What should the team change first?

A Request a much higher project quota from Google Cloud support

B Retry failed calls with exponential backoff and jitter

C Switch the dataset to a different region with spare capacity

D Grant the script's service account the BigQuery Admin role

◆ PAUSE AND DECIDE

What should the team change first?

A Raise the quota

A short-term rate limit, not a used-up quota.

B Backoff and retry

Exactly what the reason asks for.

C Another region

Moves data, and the bursts continue.

D BigQuery Admin

Permissions are not the problem.

THE SIGNAL

rateLimitExceeded on bursts: slow down, and retry with backoff.

> A scheduled query, in a project whose datasets are in europe-west2, fails with notFound for a table that clearly exists.
>
> The job was configured to run in the EU multi-region.

What should the team change?

A Recreate the table in the EU multi-region, then rerun the query

B Grant the scheduled query's account Data Viewer on the dataset

C Run the job in europe-west2, where the datasets live

D Raise the project's query quota, then retry the job later

◆ PAUSE AND DECIDE

What should the team change?

A Copy to EU

A second copy, and a residency question.

B Grant Data Viewer

Missing access would say accessDenied.

C Match the location

Jobs must run where the data lives.

D More quota

Not a quota problem.

THE SIGNAL

notFound for a table that exists: check the job's location.

> Auditors ask which people read curated.customers last month. Data Access logs are on, as BigQuery's default.
>
> An analyst with the Logs Viewer role searches, but sees no Data Access entries at all.

What should the team do?

A. Enable Data Access logs for BigQuery, then search again next month

B. Search the Admin Activity logs instead, which record every read

C. Use INFORMATION_SCHEMA.TABLE_STORAGE to list the table's readers

D. Grant the analyst Private Logs Viewer, then search the entries

◆ PAUSE AND DECIDE

What should the team do?

A Enable logs

They are already on for BigQuery.

B Admin Activity

Records changes, not reads.

C TABLE_STORAGE

Storage sizes, not readers.

D Private Logs Viewer

The role that can read Data Access entries.

THE SIGNAL

Data Access logs exist but are invisible: the reader needs Private Logs Viewer.

Module 36 in five lines

- Metrics say **something** is wrong, logs say **what**, audit logs say **who**.
- Alert on **symptoms**. Metric absence catches pipelines that go silent.
- **Aggregated sinks** to BigQuery for SQL, to Pub/Sub for outside tools.
- Admin Activity always on. **Data Access** needs Private Logs Viewer.
- Read the error **reason**: rate limit, quota, resources, access, location.

COMING UP

Next: module 37, operating streaming pipelines.